

LECTURE 7: ECONOMETRICS ACCORDING TO ED LEAMER: HAVE WE TAKEN THE CON OUT OF ECONOMETRICS? OR HAVE WE TAKEN THE TRICS OUT OF ECONOMETRICS

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OUTLINE

1. Cowles commission: mathematical statistics
2. Hendry: from general to specific
3. Leamer: “let’s take the con out of econometrics:” specification searches and sensitivity
4. Ashenfelter, Card, Angrist, Krueger, Duflo, Kremer *et al* Research Design: Randomized Controlled trials and Natural Experiments
5. Ruhm: “Shackling the Identification Police?”
6. Pearl: Directed Acyclical Graphs

HISTORY

- Active literature on methodology in the 80's, Sims, Hendry, Leamer.
Econometricians acknowledge the issues but put them aside. Had **little impact** on empirical practice.
- People come back to this from time to time.
- But: **Read Leamer!** Always interesting, always witty, and his views and insights are still very relevant.

COWLES COMMISSION

- Cowles commission: econometrics as mathematical statistics, 50's-60's, taking a particular perspective on Haavelmo. Note: lots of approaches see themselves as following in the footsteps of Haavelmo, including Cowles commission, Pearl, Heckman, but they focus on different things.
- Specify a statistical model (traditionally fully parametric, but that is not essential):

$$f(Y, X; \theta) = \varepsilon,$$

- Here Y (endogenous) and ε are K -dimensional vectors, with ε independent of X (exogenous), with a parametric distribution for ε (e.g., multivariate normal). The function $f(\cdot)$ is known, parameter of interest θ is unknown.
- Includes classic linear simultaneous equations: $\Gamma Y + XB = \varepsilon$.
- Given random sample from large population, find estimator for θ , and derive frequentist statistical properties, construct confidence interval.

COWLES COMMISSION

- Big focus on mathematical statistics
- validity of confidence intervals
- few concerns about small sample properties, rely heavily on asymptotics.
- Focus on θ , parameter of model, not necessarily on counterfactuals.

COWLES COMMISSION

- The notion is that one specifies a single model, and estimates the parameters **once**. There is no uncertainty about the model. The only uncertainty about the parameters comes from the sampling variation.
- **Applied econometricians** here are essentially **technicians/robots**. They specify the correct model, estimate it, and report the results. No need to iterate, or refine the models. A single model is used from the beginning. No role for creativity.
- In pre-analysis plans for randomized experiments this is a reasonable model of what data analysts do.
- It is unclear whether there is any role for specification testing: if we reject a model, we cannot re-estimate a different one because statistical inference is no longer valid.
- **Very far from reality in empirical work**
- Important role for **theoretical econometricians**: find new methods and derive **properties!**

LEAMER, JOURNAL OF APPLIED ECONOMETRICS, 1997

- *“The classical monk spends days copying tomes that explain how to do econometrics in Asymptotia. ... We let the classical monk determine the language that we use for discussing data analysis (t-values, standard errors), but we ignore altogether the monk’s rules for the holy sacrament of data analysis. We sin and we sin wantonly. Some of us use robots for assistance in our sins. These robots are programmed to omit variables that have low t-values and to add new variables that have high t-values. Most of us prefer to do the discarding ourselves. We analyze the data the same way that the 1000-pound gorilla makes love: ‘anyway she wants to.’ Whatever the robots or the gorillas produce, we carry the results up to the monastery of the classical monks for their blessing. We get the blessing, but only because we don’t admit that we spent the night with robots and gorillas.”*

WHAT LEAMER MEANS:

- We estimate lots of different models. We add variables, change functional forms, scratch our heads, look for new instruments, throw out units with missing data, do some tests, *etc.*
- But, then we report estimates for one model, or for a few models, with confidence intervals **as if** those were the only specifications used.
- We look at t-statistics greater than 2 and conclude that we are confident the parameter is not equal to zero.

WHITE, DATA SNOOPING, 2000

- White (2000) calls this “data snooping” and tries to construct confidence intervals that account for that.
- He assumes the econometrician specifies an algorithm, for example for selecting covariates in regression function. As long as this algorithm is fully specified we can think about taking into account the uncertainty that generates in the final estimates.
- This just kicks the can down the road. The difficulty is that we do /can not specify completely what the algorithm would be.
- Machine learning: use sample splitting to get “honest” properties. More realistic attempt to take into account uncertainty without having to fully specify analysis plan.

HENDRY: FROM GENERAL TO SPECIFIC

- Hendry acknowledges that the notion that a researcher immediately specifies the correct model and conducts inference in the classical way is unrealistic:

“The ‘conventional’ approach insists on a complete theoretical model of the phenonema of interest prior to data analysis, leaving the empirical evidence as little more than quantitative clothing.”

- He advocates a sequential, iterative approach to finding a final specification. It starts with a general model and simplifies it to a specific one:
- **General:** Build a statistical model that captures the economic model and fits the data well. Judge the statistical fit based on specification tests.
- **To Specific:** Simplify the model using additional specification test: drop covariates that are statistically insignificant.
- Check that the model still fits the data well.

- Hendry thinks about macro (time-series) models.
 - How would this work for cross-section analyses? Suppose I am interested in the relation between earnings and education.
 - Finding a statistical model that fits the data on earnings, education and individual characteristics well is not difficult. It would not convince me that the model was useful for evaluation educational policies. There are many such descriptive models that are not causal.

HENDRY: TECHNICAL POINT

- Hendry is not worried about pre-testing bias:

“The mis-specification tests are thus applied only once as statistical mis-specification tests – to the general model – so no doubts of interpretation arise. Their subsequent role in diagnostic checking during reduction does not alter their statistical properties as applied to the general model.”

- McAleer is not worried either

“Specification searching does not affect the probability of truth of a particular theory.”

WHAT TO MAKE OF HENDRY

- I don't know that that means, and agree again with Leamer:

"I can tell you that listening to words like these spoken by the mellifluous voice of McAleer's messiah, David Hendry, is soothing indeed, not because they mean anything, but only because they sound like they must. The song doesn't sound quite so soothing when the background switches from violin to digirido. In other words, pretesting matters, song or no."

HOW TO DO THE SIMPLIFICATION STEP

- Putting aside questions about inference in the final model, a big question is how to simplify the model?
- In linear regression model: iteratively drop covariates with lowest t-statistics?
Drop sets of covariates?
- Hendry develops algorithms for doing this, but unclear how they work in practice. Clearly, any algorithm can be made to work badly.
- Simplification is largely **art**, and not **science**.

LEAMER: SPECIFICATION SEARCHES AND SENSITIVITY

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- Leamer is concerned with the fact that conventional measures of uncertainty do not capture all aspects of uncertainty.
- At some level that is a concern about pretesting.
- Suppose the true model is the long regression (interest in β_1)

$$Y_i = \beta_0 + \beta_1 \cdot X_{i1} + \beta_2 \cdot X_{i2} + \varepsilon_i$$

- We first estimate this model.
 - If the t-statistic on β_2 is greater than 1.96, we report $\hat{\beta}_1$ and its ols standard error.
 - If the t-statistic on β_2 is less than 1.96 we report $\hat{\beta}_1$ and its ols standard error from the short regression

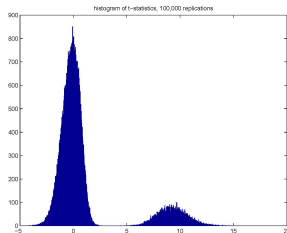
$$Y_i = \beta_0 + \beta_1 \cdot X_{i1} + \varepsilon_i$$

EXAMPLE CONTINUED

- Suppose β_1 is zero, X_{i1} and X_{i2} are both $\mathcal{N}(0, 1)$, with correlation $\rho = 0.95$, and $\varepsilon \sim \mathcal{N}(0, 1)$.
- Without the pretesting the t-statistic T on β_1 would have a normal distribution with mean zero and unit variance.
- With the pretesting: $\text{pr}(T > 1.96) = 0.13$ (instead of 0.025).

EXAMPLE CONTINUED

- Distribution of t-statistic



WHAT SHOULD WE DO INSTEAD?

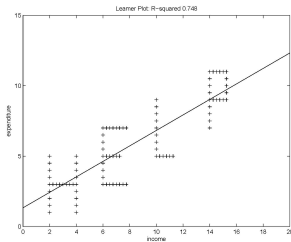
- **One Unrealistic Approach:** Specify fully what you will do under any eventuality.
- The pretesting problem above is not a difficult problem: if you specify that your strategy is to run the long regression if $|T_1| > 1.96$ and the short regression if $|T_1| \leq 1.96$, you can infer what the distribution is of the final t-statistic.
- More generally, if you fully articulate your strategy, the finite sample distribution may be difficult, but in principle is estimable, or can be approximated through resampling methods (See White, Econometrica, data snooping, 2000).

PRE-ANALYSIS PLAN

- But, that is not realistic either. Leamer argues we cannot fully articulate what we will do in all cases. (Haiti income/expenditure example).
- Maybe you look at a scatter plot, and decide based on that to try a nonlinear form rather than a linear form.
- Maybe with N_0 incomplete observations you just drop them, but with $N_1 \gg N_0$ incomplete observations you use imputation methods.
- Maybe with the first stage F -statistic larger than 20 you do TSLS, with the F -statistic less than 5 you calculate Anderson-Rubin confidence intervals.
- Too many decisions to make it realistic that you can articulate them all beforehand.

LEAMER HAITI EXAMPLE

- Consumption and Income Scatter Plot and Regression Line



LEAMER, “LET’S TAKE THE CON OUT OF ECONOMETRICS”

- *“This is a sad and decidedly unscientific state of affairs we find ourselves in. Hardly anyone takes data analyses seriously. Or perhaps more accurately, hardly anyone takes anyone else’s data analyses seriously. Like elaborately plumed birds who have long since lost the ability to procreate but not the desire, we preen and strut and display our t-values”*

LEAMER

- Leamer: be realistic about this, and systematically explore and report multiple specifications.
- In a Bayesian context, explore effect of using different prior distributions.
- Not as extreme as Manski-style bounds, but somewhat in the same spirit.
- Also somewhat in the spirit of common practice of reporting a couple of different models (put in fixed effects or not, put in interactions or not), but attempt to make this more systematic.
- Difficult, because it requires the researcher to specify (and defend) sets of specifications as capturing the relevant models.

EXAMPLE

- Suppose we are interested in the coefficient on X_i in a linear regression

$$Y_i = \beta_0 + \beta_1 \cdot X_i + \beta_2 \cdot Z_{i1} + \beta_3 \cdot Z_{i2} + \varepsilon_i$$

- We do not know whether Z_{i1} and Z_{i2} belong in the regression.
- Define $Z_i(\theta) = Z_{i1} + \theta \cdot Z_{i2}$.
- Now let us consider the least squares estimate for β_1 in the regression

$$Y_i = \beta_0 + \beta_1 \cdot X_i + \eta \cdot Z_i(\theta) + \varepsilon_i$$

for different values of θ .

EXAMPLE

- This leads to $\hat{\beta}_1(\theta)$. Report the range

$$\left[\inf_{\theta \in \Theta} \hat{\beta}_1(\theta), \sup_{\theta \in \Theta} \hat{\beta}_1(\theta) \right]$$

- This amounts to minimizing and maximizing the value of the least squares estimate for β_1 over the four specifications that include all combinations of Z_{i1} and Z_{i2} .
- In the same spirit of Rosenbaum-Rubin sensitivity analyses in program evaluation setting.
- Not clear how this generalizes to other settings. Difficult to describe range of models to be considered.

ROSENBAUM-RUBIN SENSITIVITY ANALYSIS

- The starting point is that unconfoundedness is satisfied only conditional on the observed covariates X_i and an unobserved covariate U_i :

$$Y_i(0), Y_i(1) \perp W_i \mid X_i, U_i.$$

- The idea is to specify the form and the amount of dependence of these conditional distributions on the unobserved covariate and estimate only the dependence on the observed covariate.
- Conditional on the specification of the first part the latter is typically straightforward.
- The idea is then to vary the amount of dependence of the conditional distributions on the unobserved covariate and assess how much this changes the point estimate of the average treatment effect.

ROSENBAUM-RUBIN SENSITIVITY ANALYSIS

- We specify $\Pr(U_i = 1|X_i) = \Pr(U_i = 1|X_i) = 1/2$. and

$$\Pr(W_i = 1|X_i, U_i) = \frac{\exp(\alpha_0 + \alpha'_1 X_i + \alpha_2 \cdot U_i)}{1 + \exp(\alpha_0 + \alpha'_1 X_i + \alpha_2 \cdot U_i)}.$$

- If we specify a value for α_2 , it is straightforward to estimate α_0 and α_1 .
- Similarly, we may specify

$$Y_i(w)|X_i, U_i \sim \mathcal{N}\left(\beta_{w0} + \beta'_{w1} X_i + \beta_{w2} \cdot U_i, \sigma_w^2\right).$$

- Again we specify β_2 and then estimate β_0 and β_1 using maximum likelihood methods.
- We vary the specified parameters β_2 and α_2 over some range of reasonable values and look at the corresponding range of values for τ .

SUBSEQUENT WORK

- See discussion in *Journal of Economic Perspectives*, by Angrist-Pischke, Leamer, Stock, Sims, Nevo.
- Also, paper by Deaton on randomization, and comment by Imbens.
- Design-based/Causal Literature: Instead of doing sensitivity analyses, get better data so that results are expected to be robust.
- randomized experiments are particularly robust to inclusion of covariates. Very successful in development and labor.
- Methods for robust inference: robust standard errors, semiparametric methods, nonparametric methods. Lots of theoretical work, limited success.

DESIGN-BASED/CAUSAL LITERATURE

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- Suppose we are interested in effect of binary intervention, W_i .
- Specify (short) regression model:

$$Y_i = \alpha + \tau \cdot W_i + \varepsilon_i$$

and long regression model:

$$Y_i = \alpha + \tau \cdot W_i + \beta' X_i + \varepsilon_i$$

- Randomization ensures that the correlation between W_i and X_i is small, and so estimates for τ are robust to inclusion of additional regressors X_i .

DIFFERENT VIEWS OF MERITS OF RANDOMIZATION

- Leamer: “There is therefore a sharp difference between inference from randomized experiments and inference from natural experiments.” (by natural experiments here he means what I would call observational studies)
- Deaton: “Randomized experiments cannot automatically trump other evidence, they do not occupy any special place in some hierarchy of evidence” (?)
- David Freedman (former statistician at Berkeley): “Experiments offer more reliable evidence on causation than observational studies,”
- Pearl describes himself as “a veteran skeptic of the supremacy of the RCT”

MY VIEW

- When possible, randomized experiments are more credible than observational studies (and that is why the FDA insists on them).
- When done well, often hugely influential (project Star, for evaluating effects of class size, Lalonde paper, for evaluating labor market programs, Perry preschool project, effect of preschool programs).
- But, scaling up issues are difficult. General equilibrium effects are more important in economics than in medical applications. (Interesting example in Leamer's discussion of Angrist-Pischke paper)
- Also, some questions are about specific units: not interested in average effect of breakups or mergers, but effect of break up of AT&T, or Google-Yahoo deal. In those cases experiments are not helpful.

REFERENCES

- ANGRIST, J., AND S. PISCHKE, (2010), “THE CREDIBILITY REVOLUTION IN EMPIRICAL ECONOMICS: HOW BETTER RESEARCH DESIGN IS TAKING THE CON OUT OF ECONOMETRICS,” *Journal of Economic Perspectives*, 24(2) 3-30.
- CAMPOS, J., N. ERICSSON, AND D. HENDRY, (2005), “GENERAL-TO-SPECIFIC MODELING: AN OVERVIEW AND SELECTED BIBLIOGRAPHY,” INTERNATIONAL FINANCE DISCUSSION PAPERS, NO 838, FED.
- LEAMER, E. (1978), *Specification Searches: Ad Hoc Inference with Non-experimental Data*, WILEY, NEW YORK.
- LEAMER, E. (1983), “LET’S TAKE THE CON OUT OF ECONOMETRICS,” *American Economic Review*, 73(1) 31-43.
- LEAMER, E., AND H. LEONARD (1983), “REPORTING THE FRAGILITY OF REGRESSION ESTIMATES,” *Review of Economics and Statistics*.

REFERENCES

- LEAMER, E. (1985), "SENSITIVITY ANALYSES WOULD HELP," *American Economic Review*, 75(3) 308-313.
- LEAMER, EDWARD E. "REVISITING TOBIN'S 1950 STUDY OF FOOD EXPENDITURE." *Journal of Applied Econometrics* (1997): 533-553.
- LEAMER, EDWARD E. "THINGS THAT BOTHER ME." *Economic Record* 64, NO. 187 (1988).
- PAGAN, A. (1987), "THREE ECONOMETRIC METHODOLOGIES: A CRITICAL APPRAISAL," *Journal of Economic Surveys*, VOL 1(1), 3-24.
- PEARL, J. (2000), *Causality: Models, Reasoning, and Inference*, CAMBRIDGE UNIVERSITY PRESS.
- RUHM, C, (2018) "SHACKLING THE IDENTIFICATION POLICE?" NBER WORKING PAPER
- WHITE, HALBERT. "A REALITY CHECK FOR DATA SNOOPING." *Econometrica* 68, NO. 5 (2000): 1097-1126.